# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-QUATERNION-EN · v2.0 · 2026-10-21 |
| Package | star-quaternion |
| Version | 0.1.0 |
| Capability | star.attitude.quaternion |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_quaternion |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_quaternion-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_quaternion-0.1.0-py3-none-any.whl | 58638bce93603d2df62a474c106ef84f552a367f21c3e0999607ea43e0053c39 |
| star_quaternion.py | df2c68812a8624679517765a7f2647c84156f28bd6ff20137699a75be2dcdc78 |
| star_quaternion-0.1.0-source.zip | 7b337993bfc81e0f489f80b1d6f780f92b12b8d0013e37366a8326c0c02a2ffe |

# 5. What it does, and its limits

Unit quaternion algebra, rotation matrices and axis-angle with a fixed, tested convention (scalar first, Hamilton, active rotation); agrees with NAIF SPICE and SciPy to 9e-16 and refuses non-unit quaternions and non-rotation matrices instead of normalising them silently.

No Euler angles (twelve conventions: a separate, explicit interface is needed), no interpolation, no attitude determination. Frames are the caller's: the functions rotate vectors, they do not know what the frames are called.

- Convention: quaternion (w, x, y, z) with the scalar first, Hamilton product, ACTIVE rotation; `to_dcm(q)` is the matrix R with `rotate(q, v) = R v`; `multiply(a, b)` applies b first, then a. This is the convention of NAIF SPICE (`q2m`, `m2q`, `qxq`); SciPy stores the scalar last.
- `normalize`, `conjugate`, `multiply`, `rotate`, `to_dcm`, `from_dcm`, `from_axis_angle`, `to_axis_angle`, `angle_between_deg`. `from_dcm` uses the largest-component method (exact on half-turns, where the trace method divides by zero) and returns the quaternion with w >= 0. Angles are computed with atan2: a rotation of 1e-12 degrees keeps its value.
- Measured against SPICE and SciPy on 411 cases (random rotations, identity, half-turns, 1e-9 degree rotations): matrix, matrix-to-quaternion, product, rotated vector and angle agree within 9e-16.
- Every function returns finite floats or raises `ValueError`: wrong shape, non-numeric, boolean, NaN, infinite or |value| >= 1e150 input; a zero quaternion or axis; a quaternion whose norm is further than 1e-6 from 1 where a unit quaternion is required (it is refused, not silently normalised); a matrix that is not orthonormal to 1e-9 or is a reflection.

## 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

## 7. Citation and bill of materials

```
@software{cavazzini_star_quaternion_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-quaternion}: Rotations with one stated convention},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_quaternion/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

## 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

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